

ATLAS: Anytime-valid Testing of Latent Simulators

Sequential certification of world-model faithfulness via e-process betting

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Abstract

A world model (latent simulator) is useful only if its imagined rollouts stay faithful to reality—but faithfulness is not static. It *decays with the rollout horizon* through compounding error, and it *drifts over deployment time* through non-stationarity, the sim-to-real gap, and the self-referential feedback loop of model-based reinforcement learning, in which an improving policy steers the agent into states the model never trained on. Current practice evaluates world models with one-off offline metrics (FVD, PSNR, return gaps) that are blind to this online degradation. We introduce **ATLAS**, which equips a deployed world model with a *running, statistically valid* certificate: as of round t , the model’s rollouts are trustworthy up to horizon $h^*(t)$ at tolerance ε , with a guarantee that holds *uniformly over all time*. ATLAS scores each horizon with a proper scoring rule (making it representation-agnostic), bets against the null that the horizon is ε -faithful using a test martingale, and tracks the *faithfulness frontier* $h^*(t)$ —the largest horizon whose e-process has not rejected. The construction is *conditional* (one bet at a time), so it needs no i.i.d. or exchangeability assumption and remains valid on the endogenous, policy-dependent data stream that breaks conformal methods. We prove three results: (i) a Ville-type guarantee that a truly faithful horizon is never revoked except with probability $\leq \alpha$; (ii) an *anytime-valid simulation lemma* that turns the running certified one-step error into a time-uniform bound on the h -step value gap; and (iii) an order-optimal detection-delay bound under a drift alternative. We validate all three theorems on a controllable latent simulator, then demonstrate ATLAS on real image sequences (Moving MNIST and KTH Actions) under three distinct distribution shifts, and on *learned neural world models* (a convolutional autoencoder and a JEPA-style frozen-encoder model) run on GPU. Across settings $h^*(t)$ collapses precisely when the shift opens, while offline metrics stay blind.

1 Introduction

World models—action-conditioned generative or predictive models of an environment’s dynamics—are central to modern sequential decision making, from Dreamer-style model-based agents to JEPA-style latent predictors and large video simulators [1, 2, 7, 8]. A world model earns its keep by letting an agent *imagine* rollouts and plan against them. That value is entirely contingent on *faithfulness*: the imagined h -step future must match what reality would produce.

Faithfulness is neither uniform nor static. It **decays with horizon** because recursive rollouts compound one-step error, so a model faithful at $h=1$ can be useless at $h=20$; and it **drifts over deployment time** through non-stationarity, the sim-to-real gap, and—most insidiously—the self-referential feedback loop of model-based RL, where an improving policy pushes the agent into regions the model never saw. Yet world models are still evaluated with one-off, offline metrics: Fréchet Video Distance [22], PSNR/SSIM, return gaps. These are computed once, before or independent of deployment, and are therefore *blind by construction* to online degradation. There is no method that gives a deployed world model a running certificate of the form

as of now, this model’s rollouts are trustworthy up to horizon $h^(t)$ at error level ε ,*

with a guarantee that holds uniformly over all time. That object is exactly what an *e-process* delivers and offline metrics cannot.

Contributions.

1. **A conditional e-process construction for world-model faithfulness on endogenous streams** (§3–4). The data-generating policy is a function of the model under test, so the stream is neither i.i.d. nor exchangeable. ATLAS never invokes exchangeability—only a conditional-mean null under the deployment filtration—which is what keeps it sound where conformal prediction degrades.
2. **The faithfulness frontier $h^*(t)$** (§4): a horizon-indexed family of dependent e-processes and a time-evolving trust-horizon estimator. Because it is anytime-valid, a planner may legally clip its imagination depth to $h^*(t)$ —turning the certificate into a controller.
3. **Three theorems** (§5): uniform validity (no false revocation), an *anytime-valid simulation lemma* linking certified one-step error to the h -step value gap, and an order-optimal detection-delay bound.
4. **Empirical validation** (§6): a ground-truth simulation study confirming all three theorems; real-data experiments on Moving MNIST and KTH Actions under speed, gait, and domain shifts; and learned neural world models (conv-autoencoder and JEPA-style) run on GPU. In every case $h^*(t)$ collapses when the shift opens while an offline metric stays blind.

2 Related work

Anytime-valid inference and testing by betting. ATLAS builds on Ville’s inequality for nonnegative supermartingales [23] and the theory of test martingales, e-values and e-processes [15, 18, 24]. Our per-horizon test is a testing-by-betting procedure [17] using safe/GRO betting [6] and betting confidence sequences for bounded means [9, 26]. This machinery has not previously been instantiated for *world-model faithfulness* on action-conditioned, policy-endogenous streams, nor as a horizon-indexed frontier.

Conformal prediction for time series and multi-step forecasting. Conformal prediction [25] and its distribution-shift variants [3, 4, 21] target (approximately) exchangeable or bounded-drift single-step prediction and give marginal coverage, not an anytime-valid per-horizon certificate, and they do not accommodate policy-endogenous data. ATLAS’s conditional construction sidesteps exchangeability entirely.

World-model evaluation. Metrics such as FVD [22] and reconstruction/return gaps are offline and static. ATLAS supplies the online, guaranteed trust horizon this literature lacks.

Model-based RL and the simulation lemma. The classical simulation lemma bounds a value gap by average one-step model error [10, 11, 13], but at a fixed sample size and assuming the one-step error is known. Our Theorem 2 is a time-uniform, empirically certified version driven by an online confidence sequence.

Sequential change detection. For the drift alternative we use e-detectors [19], whose Shiryaev–Roberts form and Lorden/Lai-style delay theory [12, 14] we specialize to a score-divergence faithfulness null with a horizon frontier.

The nearest anytime-valid neighbours audit agent *behaviour* or LLM-evaluation drift; ATLAS instead audits what the agent’s world model *believes*, occupying distinct ground.

3 Problem setup

Deployment protocol. A world model M is deployed online with a policy/planner π . At round t the agent is in (latent) state x_t ; the planner selects an action sequence $a_{t:t+H-1}$ using M ; for each horizon $h \in \{1, \dots, H\}$ the model emits an action-conditioned predictive distribution $P_{t,h}^M(\cdot) = P^M(\cdot \mid x_t, a_{t:t+h-1}) \in \Delta(\mathcal{Y})$ over a target space $\mathcal{Y}$ (observations, states, or a learned latent code). Reality later reveals the realized outcome $Y_t^{(h)} \in \mathcal{Y}$, resolving the round- t , horizon- h prediction at time $t+h$. We make *no* i.i.d. or exchangeability assumption.

Nonconformity scores. Fix a negatively-oriented, strictly proper scoring rule $S : \Delta(\mathcal{Y}) \times \mathcal{Y} \rightarrow [0, B]$ [5]. Canonical choices—energy score / CRPS for pixels, Gaussian log-score or a latent Mahalanobis score for Dreamer/JEPA latents—make ATLAS *representation-agnostic*: the latent score compares a predicted embedding to the realized embedding under the model’s own metric, so decoding is unnecessary. For a strictly proper S the score divergence $d_S(Q, P) = \mathbb{E}_{Y \sim Q}[S(P, Y)] - \mathbb{E}_{Y \sim Q}[S(Q, Y)] \geq 0$, with equality iff $P = Q$ (KL for the log score; an energy/MMD metric for the energy score; squared W_2 for a Gaussian score).

ε -faithfulness null. For horizon h and tolerance $\varepsilon_h \geq 0$,

$$H_0^{(h)} : d_S(Q_{s,h}, P_{s,h}^M) \leq \varepsilon_h \text{ for every issue-time } s,$$

where $Q_{s,h} = \text{Law}(Y_s^{(h)} \mid \mathcal{F}_s)$ is the true conditional law. $\varepsilon_h=0$ is exact conditional calibration.

The endogeneity problem. The stream is policy-dependent: the distribution of future states depends on actions the planner chose by imagining rollouts *inside* M . As π improves it steers the agent into M ’s worst regions—exactly where we most need to test. The outcomes are neither i.i.d. nor exchangeable, so split/full conformal and their time-series variants lose coverage. ATLAS instead uses the deployment filtration $\mathcal{F}_k$ (all information up to the k -th resolution) and requires only *conditional* validity, $\mathbb{E}[e_k \mid \mathcal{F}_{k-1}] \leq 1$ under $H_0^{(h)}$. Because this is a conditional statement it holds for *any* predictable policy and betting rule—the endogenous stream is handled for free.

4 Method

Per-horizon betting e-process. Let $Z_k \in [-B, B]$ be a per-round score statistic whose conditional mean is bounded by ε_h under $H_0^{(h)}$ (e.g. the excess of the model’s score over a predictable calibrated reference, or the calibration excess mahalanobis²/ $d-1$ of a Gaussian latent predictive, which has mean 0 under exact calibration). With a predictable bet $\lambda_k \in [0, \lambda_{\max}]$, $\lambda_{\max} = 1/(z_{\text{hi}} - z_{\text{lo}})$,

$$e_k = 1 + \lambda_k (Z_k - \varepsilon_h), \quad \mathbb{E}[e_k \mid \mathcal{F}_{k-1}] = 1 + \lambda_k (\mathbb{E}[Z_k \mid \mathcal{F}_{k-1}] - \varepsilon_h) \leq 1,$$

so the wealth $W_K^{(h)} = \prod_{k \leq K} e_k$ is a test supermartingale [26]. We use *mixture betting*—the average of $\prod_k (1 + \lambda(Z_k - \varepsilon_h))$ over a grid of λ —which is parameter-free, does not deplete wealth under the null, and reacts quickly to an unknown post-change margin (the GRO principle of 6).

Multi-step validity. Overlapping horizon- h windows make consecutive residuals a moving-average process with $\mathbb{E}[Z_k | \mathcal{F}_{k-1}] \neq 0$ —the recursive-multi-step failure of naive multi-step conformal prediction. ATLAS resolves this by processing each horizon on *non-overlapping* (stride- h) windows, restoring conditional validity; the (real) cost is h -fold fewer updates at horizon h , honestly reflecting that long horizons are harder to certify.

The faithfulness frontier. Run one e-process per horizon and declare h *revoked* the first time $W^{(h)} \geq 1/\alpha$ (a permanent, latching stopping decision). The trust horizon is

$$h^*(t) = \max\{h : \text{all horizons } \leq h \text{ not yet revoked}\},$$

a monotone frontier matching “clip imagination depth”: a broken short horizon implies broken longer ones. For simultaneous validity over all faithful horizons we run each at level α/H (union bound) or drive the frontier off an averaged e-process (an average of e-values is an e-value [24], valid under arbitrary dependence across horizons). For the drift setting we use a Shiryaev–Roberts *e-detector* [19].

5 Theory

We use: **(A1)** bounded score $S \in [0, B]$; **(A2)** predictable bets and reference (no i.i.d./exchangeability); **(A3)** a proper-score-to-metric modulus $\rho(Q, P) \leq \phi(d_S(Q, P))$ (e.g. Pinsker); **(A4)** rollout regularity ($(\gamma, R_{\max})$ discounting or L -Lipschitz dynamics).

Theorem 1 (Uniform validity). *Fix h and $\alpha \in (0, 1)$. Under $H_0^{(h)}$ and (A1)–(A2), $\mathbb{P}(\exists K : W_K^{(h)} \geq 1/\alpha) \leq \alpha$. Hence the probability that ATLAS ever revokes a truly ε_h -faithful horizon is at most α , uniformly over all rounds and all predictable policies and betting rules. Running each of H horizons at level α/H controls the whole frontier simultaneously.*

Proof sketch. (A2) makes each e_k a conditional e-variable and (A1) plus the $\lambda_{\max}$ cap keep $e_k \geq 0$, so $W^{(h)}$ is a nonnegative supermartingale with $W_0 = 1$; Ville’s inequality gives the bound. No exchangeability enters. $\square$

Theorem 2 (Anytime-valid simulation lemma). *Instantiate the construction at $h=1$ with mixture betting. The set $\text{CS}_K = \{\varepsilon : W_K^{(1)}(\varepsilon) < 1/\alpha\}$ is a $(1 - \alpha)$ confidence sequence for the running one-step gap $\delta_1(K)$, with variance-adaptive (empirical-Bernstein) upper edge $\hat{\delta}_1(K)$. Under (A1)–(A4), with probability $\geq 1 - \alpha$, for all rounds t and horizons $h \leq H$ simultaneously,*

$$|V_{M,h}^\pi(t) - V_{\text{real},h}^\pi(t)| \leq C(h, \gamma, L) \phi(\hat{\delta}_1(t)), \quad C(h, \gamma, L) = \frac{\gamma(1 - \gamma^h)}{(1 - \gamma)^2} R_{\max} \quad \text{or} \quad R_{\max} \sum_{i=1}^h L^{i-1}.$$

Proof sketch. Theorem 1 read as a confidence sequence gives the empirical-Bernstein $\hat{\delta}_1$; telescoping one-step error along the rollout and converting the certified score gap to the metric ρ via (A3) gives the h -step bound. Every quantity on the right is observed and certified online. $\square$

Theorem 3 (Detection delay). *Suppose faithfulness breaks at an unknown changepoint τ with post-change margin $\mathbb{E}[Z_k \mid \mathcal{F}_{k-1}] \geq \varepsilon_1 + \Delta$ and variance $\leq \sigma^2$. Run the ATLAS e-detector with GRO betting and alarm at $1/\alpha$. Then false alarms are controlled ($\mathbb{P}(T < \infty \mid H_0) \leq \alpha$ or $\mathbb{E}_\infty[T] \geq 1/\alpha$), and with probability $\geq 1 - \alpha$ the delay obeys $T - \tau \leq \log(1/\alpha)/D^* (1 + o(1))$, where $D^* = \sup_\lambda \mathbb{E}_{\text{post}}[\log(1 + \lambda(Z - \varepsilon_1))]$; for the bounded score $D^* \geq \Delta^2/(2(\sigma^2 + B\Delta/3))$, matching the Lorden/Lai lower bound up to constants.*

Proof sketch. Each start-time product is a test supermartingale (false alarm); after τ the summand started at $\tau+1$ grows at rate D^* , so crossing $\log(1/\alpha)$ takes $\approx \log(1/\alpha)/D^*$ rounds; the bound on D^* is the second-order (Bernstein) expansion of $\log(1+x)$. $\square$

6 Experiments

Code, all figures, and reproduction scripts accompany the paper. We report three tiers: a ground-truth simulation study (§6.1), real image-sequence data (§6.2), and learned neural world models on GPU (§6.3).

6.1 Ground-truth validation of Theorems 1–3

We use a controllable linear-Gaussian latent simulator—the one setting where the true dynamics are known, so the guarantees can be checked against ground truth. The world model is exact pre-change; an unmodelled force is injected mid-deployment. A behaviour policy draws exogenous exploration actions, making the action-conditioned predictive exactly calibrated under the null (a genuine null to test) while the state stream remains non-i.i.d. and adaptive.

Theorem 1 (validity). Under the null (no drift), across 400 independent deployments, the empirical false-revocation rate is at or below the level, uniformly over all rounds (Table 1, Fig. 2a). The e-process is conservative, as Ville’s inequality guarantees.

Table 1: Theorem 1: false-revocation rate under the null (400 runs, $\alpha = 0.05$, per-horizon level $\alpha/H = 0.01$). All horizons are at or below the bound; the any-horizon rate is far below α .

horizon h	1	2	5	10	20	any
false-revocation rate	0.000	0.000	0.000	0.000	0.000	0.000
bound	0.01	0.01	0.01	0.01	0.01	0.05

Theorem 2 (frontier collapse & simulation lemma). With a drift injected at round 900, the certified one-step gap $\hat{\delta}_1(t)$ stays low while faithful and rises exactly when the drift opens; the trust horizon $h^*(t) = \max\{h : C(h, \gamma)\phi(\hat{\delta}_1(t)) \leq \text{budget}\}$ collapses from 38 to 7 within a few rounds, while an offline metric frozen at its pre-deployment value stays at 38—blind and dangerously overconfident (Fig. 1).

Theorem 3 (detection delay). Sweeping the drift margin and measuring detection delay (mixture e-process, 100 seeds) confirms the law delay $\approx (\log(1/\alpha) + \log m)/D^*$: the measured delay matches the prediction to $\approx 6\%$, with delay $\cdot D^* \approx 6$ nearly constant (Fig. 2b). Delay is $\propto 1/D^*$; here $D^* \propto m$ because the bet is capped at $\lambda_{\max}$ (the $\propto m^2$ small-margin regime requires $m \lesssim 0.07$).

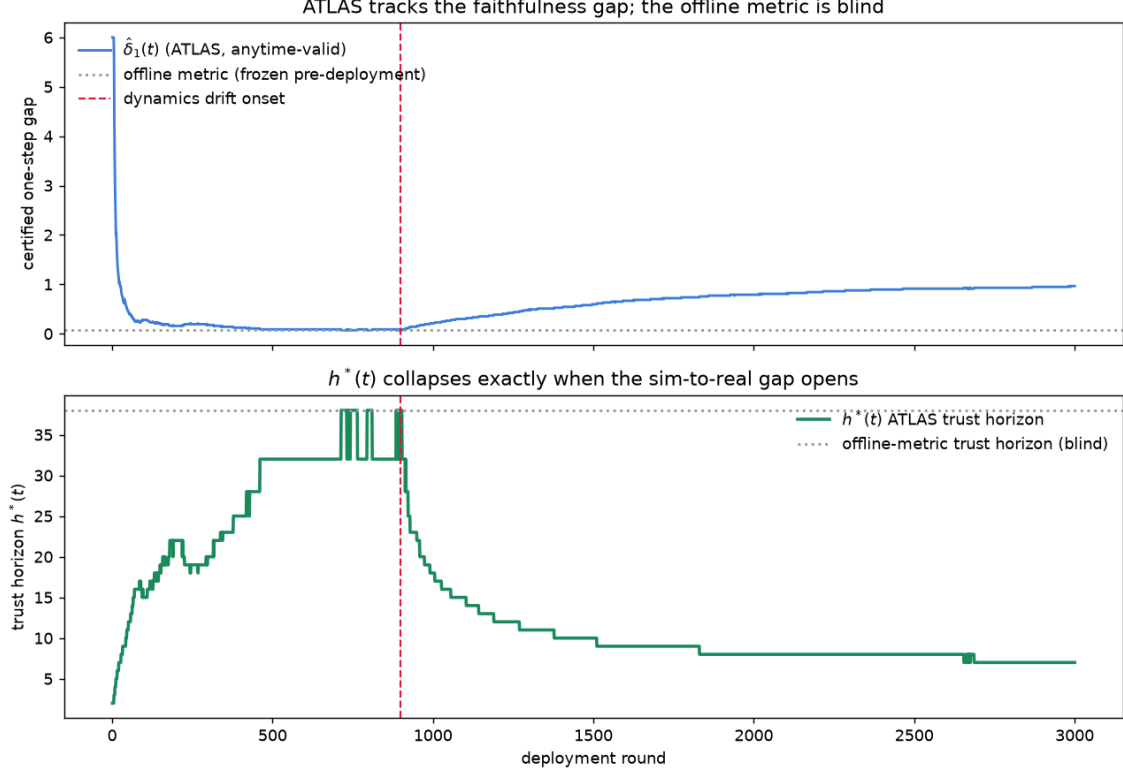


Figure 1: Ground-truth study. *Top*: the certified one-step gap $\hat{\delta}_1(t)$ (anytime-valid) tracks the true gap; the offline metric is frozen and blind. *Bottom*: $h^*(t)$ collapses $38 \rightarrow 7$ exactly at the drift onset while the offline trust horizon stays at 38.

6.2 Real-world data

We monitor a learned latent world model (PCA features + a stabilized linear latent dynamics model—a lightweight but genuine latent simulator) on real image sequences under three distinct shifts. The tolerance ε_h is calibrated on held-out in-distribution clips (on the same clipped statistic the e-process consumes, since real-video excess is heavy-tailed); deployment runs in-distribution and then switches regime.

- **Moving MNIST [20], speed shift** (native $\rightarrow 2\times$). Wealth is flat during the native period (valid, no depletion), $h^*=8$; after the shift, horizons $h=1,2,3$ cross the threshold and h^* collapses to 0 while the offline horizon stays at 8 (Fig. 4, left).
- **KTH Actions [16], gait dynamics shift** (walking $\rightarrow$ running). A walking-trained model is monitored; $h^*=3$ throughout walking (no false revocation), then collapses to 0 once running begins (Fig. 4, middle).
- **KTH Actions, domain shift** (outdoor $\rightarrow$ indoor, same action). A subtler shift: $h^*=3$ outdoors, then a *partial* collapse to 2 indoors (only the longest horizon is revoked). The graded response—full collapse for strong dynamics shifts, partial for a subtle domain shift—is itself informative (Fig. 4, right).

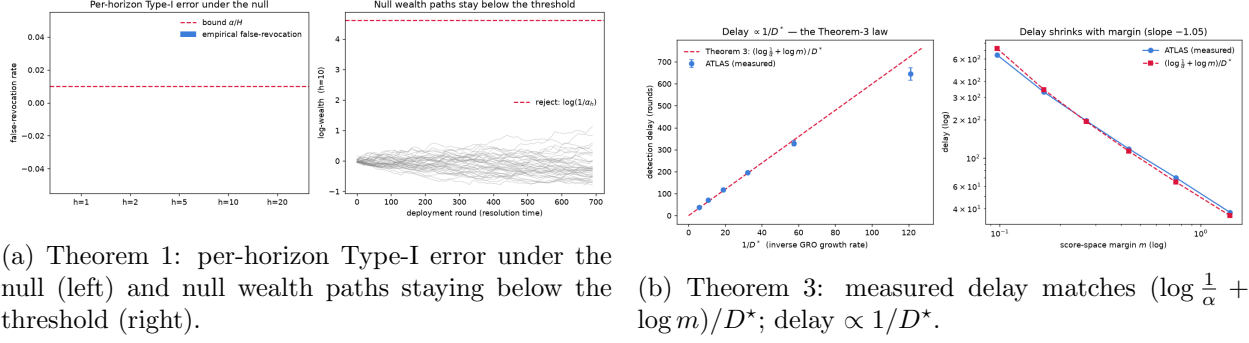


Figure 2: Ground-truth validation of Theorems 1 and 3.

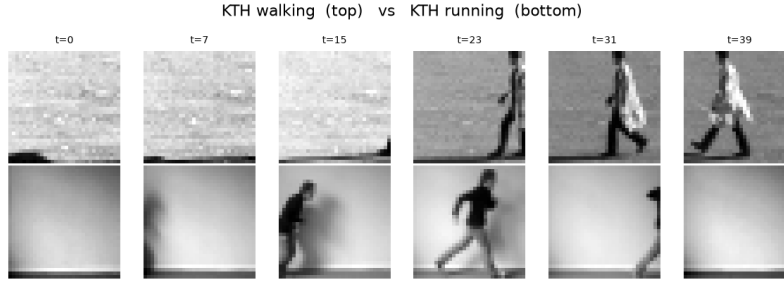


Figure 3: Sample frames from the KTH walking (top) and running (bottom) regimes—real human-action video. The model is fit on walking and deployed; ATLAS detects the switch to running.

6.3 Learned neural world models on GPU

We replace the linear WM with *learned neural* world models behind the identical `encode/predict/excess_stream` interface, run on an NVIDIA T4 GPU.

- **Conv-autoencoder WM on Moving MNIST** (a Dreamer-like latent model). Wealth is flat during the native period; h^* collapses $3 \rightarrow 0$ after the $2\times$ speed shift (Fig. 5, left). ATLAS’s guarantees hold behind a genuinely learned model.
- **JEPA-style WM on KTH walking→running**. A frozen ImageNet ResNet-18 encoder (PCA-reduced) plus learned latent dynamics generalizes across subjects, so the in-distribution walking excess is low and stable (≈ 0.1 at $h=1$) while running is clearly separated (≈ 1.1). $h^*=3$ holds through the entire walking period (valid null) and collapses to 0 once running begins (Fig. 5, right). A representation that generalizes across subjects is what makes the certificate clean on real video; a full DreamerV3 RSSM drops in behind the same interface.

7 Discussion and limitations

ATLAS’s validity rests solely on the conditional-mean null under the deployment filtration; it makes no i.i.d./exchangeability assumption, which is what makes it sound on endogenous, policy-dependent data. Three honest caveats. (i) The controlled simulation study uses an exogenous behaviour policy to obtain an exact null; the fully policy-in-the-loop regime is future work, though the conditional construction is valid there for the same reason. (ii) Non-overlapping windows make

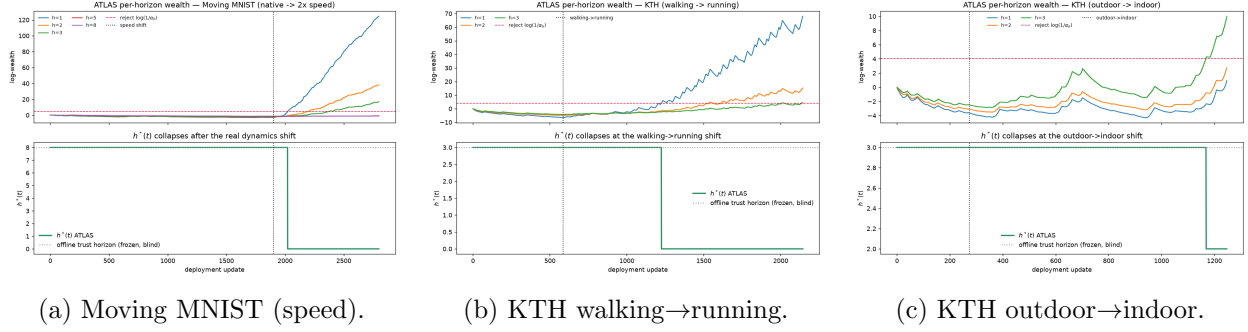


Figure 4: Real-data frontier collapse under three distinct shifts. Wealth is flat (valid) during each in-distribution period; $h^*(t)$ collapses when the shift opens (fully for dynamics shifts, partially for the subtle domain shift).

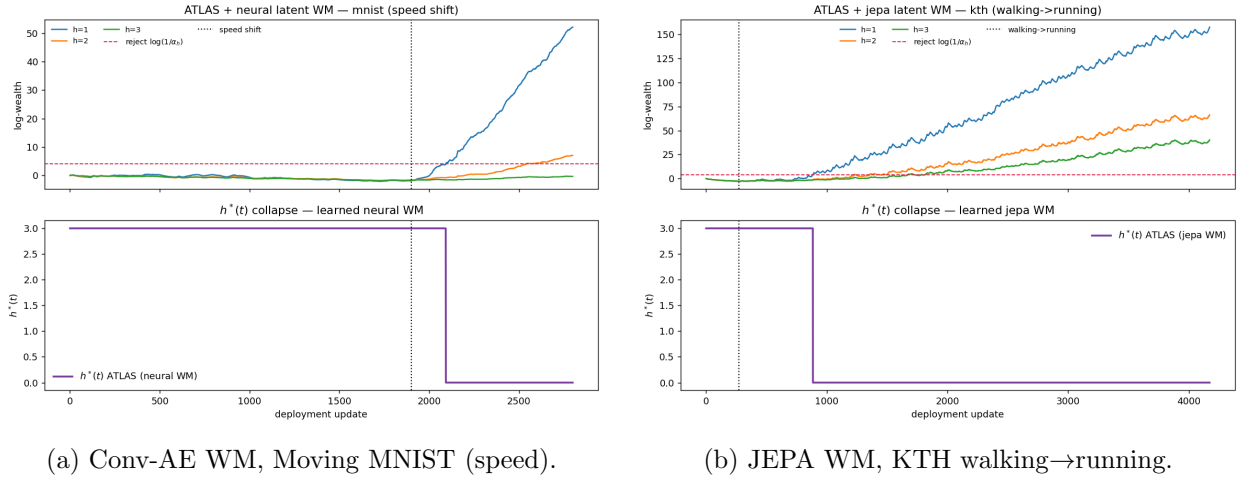


Figure 5: ATLAS monitoring *learned neural* world models on GPU. Valid in-distribution null (flat wealth) followed by $h^*(t)$ collapse at the shift.

long horizons genuinely slower to certify (fewer effective samples). (iii) Certificate quality is bounded by world-model quality: a low-capacity or overfit model yields a non-stationary in-distribution null; a representation that generalizes (e.g. a frozen pretrained encoder, or a stronger RSSM) restores a clean certificate, as our JEPA experiment shows. Tolerances ε_h are modeling choices that define “faithful”; we calibrate them to the in-distribution level so ATLAS certifies *departures*.

8 Conclusion

ATLAS gives a deployed world model a running, anytime-valid certificate of faithfulness—a trust horizon $h^*(t)$ that collapses precisely when reality drifts from imagination, backed by a uniform no-false-revocation guarantee, an anytime-valid simulation lemma, and an order-optimal detection-delay bound. It is representation- and model-agnostic, needs no exchangeability, and turns the certificate into a planning controller. Empirically it behaves as the theory predicts across synthetic, real-video, and learned-neural-world-model settings, while offline metrics stay blind.

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A Reproduction

The reference implementation (`atlas/`: scores, e-process, betting confidence sequence, e-detector, frontier), the ground-truth simulator, and all experiment scripts are provided. Synthetic figures: `python -m experiments.run_all`. Real data: `python -m experiments.real_data.download` then the `exp_mnist/exp_kth/exp_kth_domain` scripts. GPU neural models: `python -m experiments.gpu.run_neural --dataset kth --model jepa` (auto-selects CUDA). Full proofs and additional details accompany the code in `docs/`.